

SELECTED WORK

Engineering Portfolio

Mechanical design, hands-on fabrication, manufacturing, and sensor evaluation

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Mechanical Engineering student at the George Washington University
Bachelor of Science expected May 2029 | Beverly, Massachusetts

I design mechanical assemblies, build components in the machine shop, and work with experimental data. This portfolio presents my contributions to an electric boat propulsion system, production engineering at Flowserve, and stereo-camera and inertial-sensor research in GW's Estimation, Perception & Intelligent Computing Lab.

01 Electric boat drivetrain and mounts

SolidWorks modeling, structural analysis, component fabrication, and propulsion-system assembly.

02 Manufacturing engineering at Flowserve

Production work instructions, equipment traceability, drawing interpretation, and facility planning.

03 Visual inertial odometry research

Linux setup, sensor-data collection, and comparison of two motion-estimation pipelines.

Tools and qualifications

SolidWorks and Onshape | FEA | Engineering drawings | Machining and mechanical assembly | Excel
| Python and C++ | Linux and Git

Certified SOLIDWORKS Associate | Lean Six Sigma Yellow Belt

01 MECHANICAL DESIGN AND FABRICATION

Electric Boat Propulsion

GW Electric Yacht Club | ASNE Promoting Electric Propulsion Competition | December 2025 - Present

40+

hours of machine-shop fabrication

ME1616

Motenergy propulsion motor

The engineering challenge

Develop a drivetrain and motor mounting system that transfers power from an electric motor to the propeller while integrating the propulsion and steering hardware into the boat. The project combined mechanical design with fabrication and assembly under a compressed competition schedule.

My contribution

I designed the drivetrain and motor mounting system in SolidWorks. I modeled the motor using drawings and physical measurements, developed the assembly, and produced engineering drawings. I evaluated the motor mount with finite element analysis under expected propulsion and impact loads.

From CAD to hardware

I led more than 40 hours of machine-shop fabrication, making transom mounts, brackets, supports, and structural plates. I worked with the team to assemble, test, and troubleshoot the propulsion and steering systems. I also reverse-engineered the front cowl geometry for a design intended for 3D printing.



Functional power-transfer schematic. Conceptual only; not a CAD drawing or fabrication specification.

What I learned

This project connected my CAD work to physical interfaces, component fit, fabrication, and assembly. It gave me practical experience carrying a mechanical design into hardware and working through integration problems with a team.

02 PRODUCTION ENGINEERING

Manufacturing at Flowserve

Manufacturing Engineer Intern | Ipswich, Massachusetts | May - August 2026

500+

lifting devices documented

\$60,000+

in proposed facility and equipment upgrades

The engineering challenge

Support a live manufacturing operation through clearer production instructions, better equipment records, and practical facility planning. My work involved engineering, maintenance, EHS, and operations teams.

Production procedures

I authored and revised procedures for painting, rust prevention, shipping, sanding, and smear testing. An approved smear-test work instruction was implemented in production. This work taught me to translate a shop-floor process into instructions that operators can follow consistently.

Drawings and manufacturing support

I interpreted industrial valve drawings to determine custom shipping requirements and supported evaluation of test-bay enclosures for pressure capacity and safety requirements. These tasks connected engineering information to decisions about material handling and the production environment.

Data and equipment traceability

I used Excel to analyze paint-booth emissions and air-quality data for VOC tracking and environmental reporting. I documented more than 500 lifting devices and created standardized identification labels and safety materials to improve equipment tracking and inspection readiness.

Facility and equipment planning

I collaborated with maintenance, EHS, and operations on CAPEX proposals and vendor reviews for more than \$60,000 in potential upgrades. My contribution was proposal development and evaluation; the amount represents planned improvements rather than measured savings.

What I learned

Manufacturing engineering depends on documentation, usable data, and communication with the people operating and maintaining equipment. I learned to connect technical work to practical production needs.

03 EXPERIMENTAL EVALUATION

Camera and IMU Research

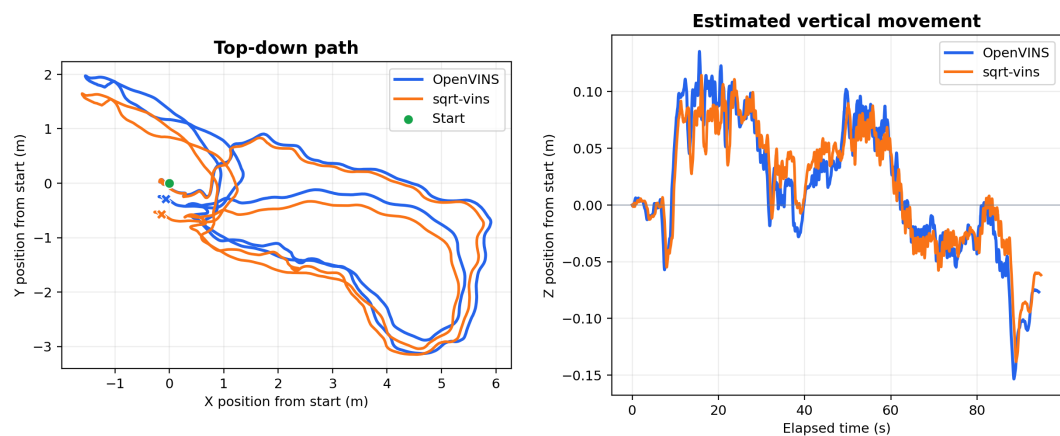
Undergraduate Researcher | GW Estimation, Perception & Intelligent Computing Lab | August 2026 - Present

The engineering challenge

Evaluate motion estimates from an EGO1GS stereo-camera and inertial measurement unit system. Visual inertial odometry combines image features and motion-sensor measurements to estimate position and orientation.

My contribution

I configured a Linux environment with Python, C++, and GitHub repositories, collected and reviewed MCAP sensor recordings using Foxglove, and worked on the calibration and VIO workflow. Both estimators processed the same walking recording.



Actual trajectory figure from my September 2026 comparison report. Paths are translated to start at the origin.

Metric	sqrt-vins	OpenVINS
Estimated poses	2,846	2,831
Estimated path length	39.77 m	40.50 m
Processing time	1 min 51 s	2 min 25 s
End-to-start distance	0.59 m	0.31 m

Results and limits

The trajectories differed by **11.6 cm in cross-estimator ATE**. sqrt-vins processed the recording 34 seconds faster; OpenVINS ended closer to the start. One roughly 94-second trial without external ground truth measures agreement, not absolute accuracy.